Calibrated INT8 Quantization: Efficient EuroSAT Inference with Minimal Accuracy Loss

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Abstract. We optimize a supplied MobileNetV2 student for satellite land-use classification using static INT8 quantization. Per-channel weights and 99.99th-percentile calibration deliver our best verified accuracy–efficiency balance without retraining. The selected model, Q3a, retains 99.76% of baseline accuracy and achieves a 4.29× speedup in Qualcomm AI Hub's proxy-device profile. We select it over a smaller pruned model whose accuracy did not recover within the available training budget.

98.92%	70.7% smaller	3.06× faster	67.7% less energy
Overall accuracy	8.506 → 2.491 MiB	2.554 → 0.836 ms	CPU estimate per image

1. The task and our test setup

The supplied student classifies EuroSAT RGB imagery into ten land-use classes. We preserve the required FP32 input named `input`, shape $(1, 3, 224, 224)$, and ten output scores. Images are resized to 224×224, converted to tensors, and normalized with the supplied ImageNet mean and standard deviation.

Following the judge instruction recorded in the project, final accuracy is evaluated on all 27,000 images, including calibration images. Consequently, it is a full-dataset competition measurement, not an independent held-out generalization estimate. All baseline numbers here are our local measurements; organizer-provided `baseline_metrics.json` was unavailable.

We export the unchanged checkpoint with ONNX opset 13 and `dynamo=False`, and use ONNX Runtime 1.31.0 with its CPU execution provider. Export-verification code compares real-image logits and predicted classes with PyTorch. Using ONNX for both baseline and final comparisons avoids attributing a runtime change to compression.

Local measurements use an 11-core Apple M3 Pro, 18 GB RAM, mains power, batch one, and ONNX Runtime's default thread setting. The milestone alternates candidates over five rounds; each round uses ten warm-ups, 100 timed inferences, and a 15-second energy run. Table 2 uses the median across rounds, with median inference latency within each round. RAM is sampled process RSS, not model storage.

2. How we chose the winning model

Our pipeline is checkpoint → FP32 ONNX → graph preprocessing → calibration → static INT8 QDQ → validation. Preprocessing performs shape inference and graph optimization before quantization. QDQ represents quantization explicitly with QuantizeLinear and DequantizeLinear nodes; static ranges are calibrated before deployment [1].

The principal calibration set contains 300 fixed images, 30 per class, sampled with seed 42. We compare per-tensor and per-channel weight scales, signed/unsigned activation types, MinMax, entropy and percentile calibration, and selective FP32 retention. The selected Q3a configuration uses signed INT8 weights and activations (S8S8), per-channel weights, 99.99th-percentile calibration, and no excluded layers. No retraining is needed for this selected artifact.

Table 1: Quantization study on all 27,000 images. Size uses 2^{20} bytes (MiB).

Configuration	Acc. (%)	MiB
Original FP32	99.16	8.506
Q0: per-tensor, MinMax, U8S8	96.32	2.277
Q1: per-channel, MinMax, S8S8	98.09	2.491
Q1b: Q1 with U8S8	98.09	2.491
Q2: per-channel, entropy	98.09	2.491
Q3a: percentile 99.99	**98.92**	**2.491**
Q3b: percentile 99.999	98.69	2.491
Q4: Q3a, first/last layers FP32	98.86	2.528
Q6: 100 / 1,000 calibration images	98.83 / 98.90	2.491
Q7: FP16	99.17	4.285

Selection rationale. Q1b versus Q0 isolates weight-scale granularity at U8S8 and improves accuracy by 1.77 percentage points. Q3a then gains 0.83 points over per-channel MinMax. Increasing the percentile to 99.999 retains more extreme values but performs worse here. Protecting the first and last layers also fails to improve accuracy. The 100- and 1,000-image checks remain within 0.09 points of the 300-image result; this supports that choice in this experiment, without establishing general convergence.

Q3a meets our targets of at least 98.9% overall accuracy, about 95% on PermanentCrop, and at most 2.5 MiB. FP16 preserves accuracy but is larger and slightly slower than FP32 in our repeated CPU benchmark (2.708 versus 2.554 ms). Cross-layer equalization [2], adaptive rounding and quantization-aware training were considered but are not claimed as completed experiments.

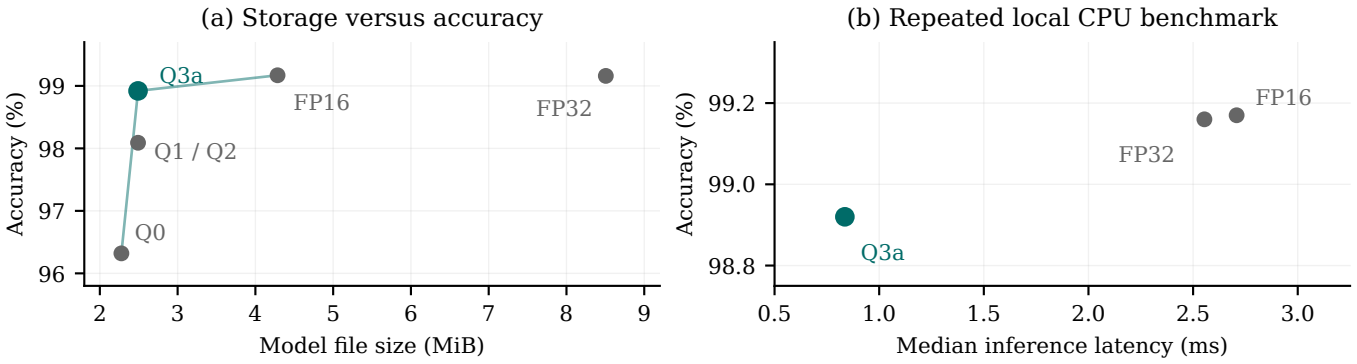


Figure 1: Quantization tradeoffs on all 27,000 EuroSAT images. (a) Accuracy versus storage for representative candidates; the line joins selected tradeoff points. (b) The three finalists measured in the same five-round local benchmark. Q3a is highlighted: 98.92% accuracy at 2.491 MiB and 0.836 ms per image.

3. What the winning model achieved

Table 2: Selected model versus identical-runtime baseline.

Metric	FP32	Q3a INT8
Overall accuracy (%)	99.16	98.92
Worst class: PermanentCrop (%)	95.84	95.24
File size (MiB)	8.506	2.491
Parameters (M) / MACs (G)	2.220 / 0.2995	same
Local median latency (ms)	2.554	0.836
Local peak RSS (MiB)	87.91	76.86
Estimated energy (J/image)	0.05993	0.01938
AI Hub estimated latency (ms)	26.690	6.225
AI Hub estimated peak (MiB)	28.10	29.35

Q3a reduces storage by 70.7% (3.41× smaller) and median local latency by 67.3% (3.06× faster), while losing only 0.24 percentage points of accuracy. It retains 99.76% of baseline accuracy; no class loses more than 0.95 points. Local peak RSS falls 12.6%. Parameters and MACs remain unchanged: these gains come from lower-precision representation and execution, not structural removal. Estimated energy falls 67.7%; CodeCarbon uses a CPU-load/TDP estimate rather than power sensors, with variation across rounds.

Device choice. We used the Dragonwing RB3 Gen 2 Vision Kit (QCS6490, Qualcomm Linux 1.6). The project catalog check found no UNO Q / QRB2210 profile and selected this lower-tier listed Linux IoT alternative. Both saved ONNX profiles execute entirely on CPU. AI Hub estimates a 4.29× speedup, while peak memory rises 4.5%. These are proxy-device results, not UNO Q measurements or evidence of NPU acceleration.

4. Pruning + INT8: lessons for next time

We removed blocks 6, 9 and 16, trained for one epoch with teacher distillation on 5,400 images, and recalibrated INT8. The extra compression cost too much accuracy:

Comparison	Q3a (chosen)	Pruned + INT8
Accuracy (%)	98.92	96.52
PermanentCrop (%)	95.24	89.28
Model size (MiB)	2.491	2.052

What went wrong. One recovery epoch did not repair the pruning damage: on the same 2,700-image subset, accuracy fell from 97.56% after pruning/BN adjustment to 96.00% after training. Full-dataset accuracy was 96.49% before INT8 and 96.52% after it, placing the observed deficit before quantization. One epoch consumed 850 seconds (14.2 minutes). We lacked time to tune the learning rate, compare teacher-assisted versus label-only recovery, or complete the separate evaluation excluding fine-tuning images. These results do not establish whether longer recovery would succeed.

Next steps. Start with fewer removed blocks or milder channel cuts; compare lower learning rates and longer recovery schedules, saving the best validation checkpoint. Test distillation against label-only training, then recalibrate and quantize only recovered candidates. Reserve time for disjoint evaluation and repeated latency/energy measurements. Q3a remains the verified reference to beat.

5. Reproducibility and conclusion

The selected `student_model_int8.onnx` matches Q3a's SHA-256 (prefix `3335203487330816`). Rebuild from the supplied checkpoint with `export_onnx.py`, then `quantize.py` using `-per-channel`, `-calib-method percentile`, `-percentile 99.99`, and `-types s8s8`. Fixed calibration indices and saved settings identify the exact inputs. Use `check_quant.py`, `measure.py`, and `timing.py` for validation; pruning configurations are saved separately. AI Hub profile IDs are `jgk68x4n5` (FP32) and `j5wy46y4g` (INT8). For this task and budget, per-channel percentile-calibrated INT8 provides our best verified balance of accuracy, storage and deployment efficiency. The more aggressive pruning extension is retained as evidence for future recovery experiments, rather than selected for submission.

References

[1] ONNX Runtime. *Quantize ONNX models*. Documentation, accessed October 2026.

[2] M. Nagel et al. Data-Free Quantization Through Weight Equalization and Bias Correction. *ICCV*, 2019, pp. 1325–1334.

Appendix A. Dashboard data at a glance

Recorded experiments from 13 October 2026. Values below come directly from the JSON files behind the dashboard; archived and smoke-test runs are excluded. Teal highlights the submitted model. The first two pages remain the main technical report.

A1. Full-dataset experiments: all 27,000 EuroSAT images

Run	Acc. %	Worst %	Params M	MACs G	Size MiB	Latency ms	RAM MiB	Energy J/image	Time basis
FP32 ONNX	99.16	95.84	2.220	0.2995	8.506	2.554	87.91	0.05993	M
FP32 PyTorch	99.16	95.84	2.237	0.3262	8.772	19.862	303.19	0.32409	S
Q0	96.32	88.13	2.220	0.2995	2.277	0.905	75.56	0.02251	P
Q1	98.09	95.45	2.220	0.2995	2.491	0.824	76.42	0.01843	P
Q1b	98.09	95.45	2.220	0.2995	2.491	0.928	76.42	0.05401	P
Q2	98.09	95.45	2.220	0.2995	2.491	0.835	77.17	0.02076	P
Q3a (selected)	**98.92**	**95.24**	**2.220**	**0.2995**	**2.491**	**0.836**	**76.86**	**0.01938**	**M**
Q3b	98.69	95.92	2.220	0.2995	2.491	0.836	77.08	0.01809	P
Q4	98.86	94.88	2.220	0.2995	2.528	1.056	77.84	0.02448	P
Q6a: 100 images	98.83	94.72	2.220	0.2995	2.491	0.912	77.16	0.11928	P
Q6b: 1,000 images	98.90	94.96	2.220	0.2995	2.491	0.838	78.52	0.03317	P
Q7: FP16	99.17	95.88	2.220	0.2995	4.285	2.708	89.33	0.06280	M
B3: pruned FP32	96.49	88.52	1.834	0.2627	7.028	2.263	83.25	0.06386	P
B3: pruned INT8	96.52	89.28	1.834	0.2627	2.052	0.740	75.22	0.01599	P

Reading the table. M = median of five milestone rounds; P = provisional single-session measurement; S = single-session reference. Latency is the median within a round. All energy values are CodeCarbon estimates; unusually high provisional values should not be treated as settled comparisons. Worst class is Forest for Q0, Pasture for Q1/Q1b/Q2, and PermanentCrop otherwise. ONNX and PyTorch counts/runtime overhead differ; compare within the same runtime. Q0–Q7 settings are defined in Table 1; B3 removes blocks 6, 9 and 16 and adds one recovery epoch.

A2. Pruning exploration: 2,700-image quick evaluation, before recovery training

All candidates use adaptive BN before evaluation. Counts are measured with the same PyTorch/thop method as the reference. These subset accuracies are not directly comparable with A1; latency, RAM and energy were not measured for these Stage A runs.

Candidate	Acc. %	Worst %	Params M	MACs G	MiB
Reference + BN	99.37	97.60	2.237	0.3262	8.506
Drop block 3	97.33	91.20	2.228	0.2966	8.472
Drop block 5	98.52	93.60	2.222	0.3139	8.449
Drop block 6	98.96	95.60	2.222	0.3139	8.449
Drop block 8	98.67	96.00	2.182	0.3153	8.300
Drop block 9	98.96	95.20	2.182	0.3153	8.300
Drop block 10	98.48	93.20	2.182	0.3153	8.300
Drop block 12	98.48	93.20	2.118	0.3025	8.058
Drop block 13	98.41	95.60	2.118	0.3025	8.058
Drop block 15	98.44	94.00	1.917	0.3103	7.291
Drop block 16	98.93	96.00	1.917	0.3103	7.291
Stack 1 block	98.96	95.60	2.222	0.3139	8.449
Stack 2 blocks	98.52	93.20	2.168	0.3030	8.243
Stack 3 blocks	97.56	92.00	1.848	0.2871	7.028
Stack 4 blocks	94.07	84.40	1.793	0.2761	6.822
Stack 5 blocks	91.07	79.60	1.778	0.2638	6.764

Candidate	Acc. %	Worst %	Params M	MACs G	MiB
Global 10%	77.19	47.20	1.808	0.2247	6.878
Global 20%	57.19	16.67	1.439	0.1789	5.476
Global 30%	39.41	5.67	1.095	0.1421	4.170
Global 40%	18.85	0.00	0.828	0.1099	3.157
Uniform 10%	76.26	43.20	1.784	0.2366	6.788
Uniform 20%	65.00	51.20	1.421	0.1984	5.409
Uniform 30%	39.15	27.00	1.092	0.1527	4.158
Uniform 40%	21.37	0.00	0.810	0.1144	3.089
30% BN-scale	15.81	0.00	1.025	0.1584	3.903
30% Taylor	22.74	0.00	1.003	0.1556	3.819

Depth-stack key. Blocks are added in this order: 6, 9, 16, 8, 5. Thus "Stack 3" removes 6+9+16; "Stack 4" also removes block 8.

Width settings. Global and uniform sweeps use magnitude importance. The requested percentage is a channel-pruning setting, not a percentage of total weights. BN-scale and Taylor are alternative importance criteria at 30%.

Why Q3a was selected. It satisfies the accuracy and storage targets without recovery training. The smaller B3 INT8 candidate loses 2.40 accuracy points and misses the worst-class target.

Source and precision. A1 uses `results/*.json`, replacing timing fields with `milestone_timing` when present, as the dashboard does. A2 uses `results/stageA/*.json`. Figures are rounded only for display; the saved JSON records retain the measurement precision. M = million parameters; G = billion multiply–accumulate operations; MiB = 2^{20} bytes.